Financial Time Series Analysis

Lecture 8

- Homework #3 due Friday, April 16, 1:30pm
- TA Session Friday April 16, 2021, 11:00am - 12:00pm (EDT)
- TA Session Monday April 19, 2021, 7:30pm - 8:30pm (EDT)

Outline

- Transforming from non-stationarity to stationarity
  - Seasonality
  - Differencing
- Forecasting Example
- ARCH/GARCH

Non-Stationary to Stationary

- Most series in financial economics and finance are not stationary. Since our methods are for stationary processes, we want to transform the original data set into one that is much closer to stationarity.
- There are several standard methods in common use to accomplish this including:
  - Note if the dataset exhibits unequal variance (other than volatility clustering), for example larger variance for higher values. Consider a Box-Cox transformation.
  - If the time series has a periodically repeating pattern, then remove a seasonal component
  - If the time series has a smooth upward or downward behavior, fit and remove a trend component
  - After removing trends and seasonal components, if the resulting series is not weakly stationary, consider differencing

Trend and Seasonal Components: 5

- We next consider the case in which there is a seasonal (periodic) component with period d present.
- First, we define indicator functions,

$$I_j(t) = \begin{cases} 1 & \text{if t mod d = j} \\ 0 & \text{otherwise} \end{cases}$$

  for $1 \leq j \leq d$.
- Second, we consider a model for the trend, for example a linear trend, $m_t = \beta_0 + \beta_1 t$. Thus our model becomes

$$X_t = \beta_0 + \beta_1 t + \sum_{j=2}^{d} \beta_j I_j(t),$$

  and we estimate the $d+1$ parameters $\{\beta_i,\ i = 1, \ldots, d\}$ using least squares by minimizing $\sum (X_t - \hat{X}_t)^2$.

Trend and Seasonal Components: 6

- Note that one of the indicators is omitted as the sum of all indicators must be 0. Define $\{\omega_j,\ 1 \le j \le d\}$ by $\omega_1 = 0, \omega_2 = \hat{\beta}_2, \ldots, \omega_d = \hat{\beta}_d$. We need to adjust the seasonal components so that they sum to 0, consequently, we define $\bar{\omega} = \frac{1}{d}\sum_{j=2}^{d} \hat{\beta}_j$. The seasonal components are then given by $\hat{s}_j = \omega_j - \bar{\omega},\ 1 \le j \le d$, with $\hat{s}_t = \hat{s}_{t-d}$ for $t > d$. The trend estimate is

$$\hat{m}_t = \hat{\beta}_0 + \hat{\beta}_1 t + \bar{\omega}.$$

  The same ideas can be applied for more complicated trend models.

Differencing: 1

- We define the differencing operator ∇ by $\nabla X_t = X_t - X_{t-1} = (1-B)X_t$. Note that one can create polynomials in ∇ and B for example:

$$\nabla^2 X_t = (1-B)^2 X_t = (1-2B+B^2)X_t = X_t - 2X_{t-1} + X_{t-2}.$$

- Differencing can remove polynomial trends. Consider the linear and quadratic cases. Suppose $m_t = \beta_0 + \beta_1 t$, that is $X_t = \beta_0 + \beta_1 t + Y_t$, where Y_t is stationary.

$$\begin{aligned}
\nabla X_t &= X_t - X_{t-1} \\
&= \beta_0 + \beta_1 t + Y_t - \beta_0 - \beta_1(t-1) - Y_{t-1} \\
&= \beta_1 + Y_t - Y_{t-1} \\
&= \beta_1 + \text{ a stationary process}
\end{aligned}$$

Differencing: 2

Similarly, suppose instead there is a quadratic term, i.e. $X_t = \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t$. Applying differencing twice, $(1-B)^2 = 1 - 2B + B^2$, to the time series, we find

$$
\begin{aligned}
(1-B)^2 X_t &= X_t - 2X_{t-1} + X_{t-2} \\
&= \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t \\
&- 2(\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2 + Y_{t-1}) \\
&+ \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 + Y_{t-2} \\
&= 2\beta_2 + (Y_t - 2Y_{t-1} + Y_{t-2}) \\
&= 2\beta_2 + \text{ a stationary process}
\end{aligned}
$$

Differencing: 3

- One can also use differencing to remove both a seasonal and a linear trend. Suppose that $X_t = \beta_0 + \beta_1 t + s_t + \epsilon_t$, and the seasonal term has period d, i.e. $s_t = s_{t-d}$.
- In this case the differencing should be done over d time periods rather than simply one time period using ∇. Instead we define a "d-lagged difference operator"

$$\nabla_d = X_t - X_{t-d} = (1 - B^d) \neq \nabla^d.$$

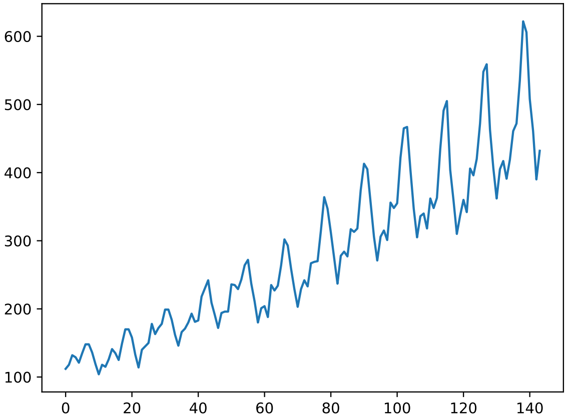
Differencing: 4

Applying this operator to the time series, we find

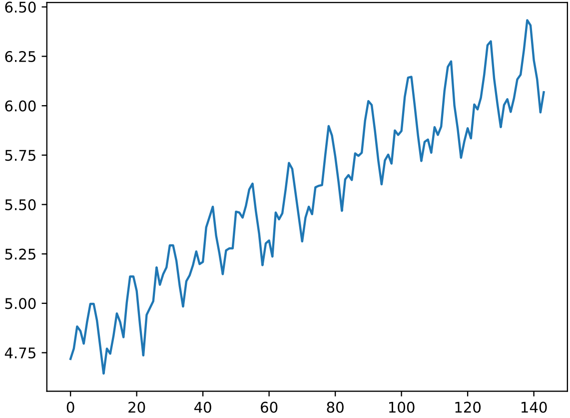
$$
\begin{aligned}
\nabla_d X_t &= \nabla_d m_t + \nabla_d s_t + \nabla_d \epsilon_t \\
&= m_t - m_{t-d} + s_t - s_{t-d} + \epsilon_t - \epsilon_{t-d} \\
&= \beta_1(t - (t - d)) + 0 + \epsilon_t - \epsilon_{t-d} \\
&= \beta_1 d + \text{ a stationary process.}
\end{aligned}
$$

Thus this differencing operator has transformed the original time series into a weakly stationary process.

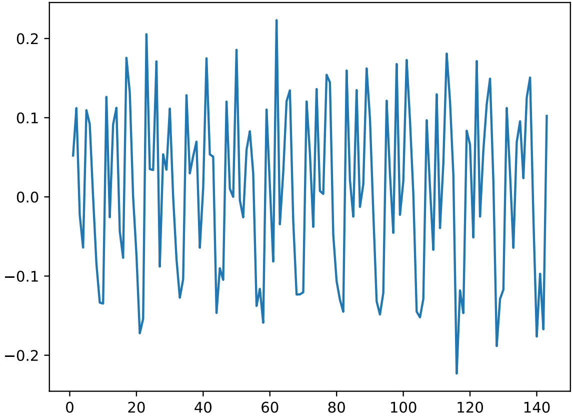
International Airline Passengers, 1949-1960



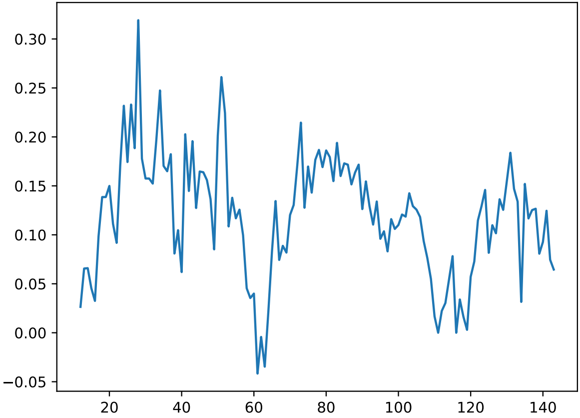
Log(Int. Airline Passengers): 1949-1960



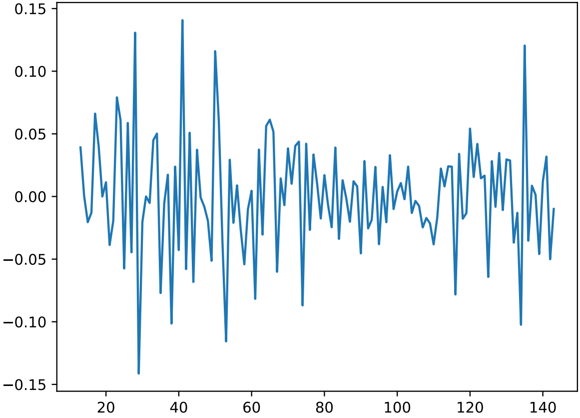
Difference(Log(Airline Passengers))



Seasonal Adj. (12) Log(Airline)



Diff(Seasonal12Log(Airline))



Forecasting with PMDARIMA: 1

```
import numpy as np
import matplotlib.pyplot as plt
import pmdarima as pm
from pmdarima.model_selection import train_test_split
# Load/split data
y = pm.datasets.load_airpassengers()
y = np.log(y)
train, test = train_test_split(y, train_size=100)
# Fit the model
model = pm.auto_arima(train, seasonal=True, m=12)
fit1 = pm.auto_arima(y, m=12, trace=True,
suppress_warnings=True)
Best model: ARIMA(2,1,1)(0,1,0)[12] model.summary()
```

Forecasting with PMDARIMA (100): 2

	coef	std err	z	P>|z|	[0.025	0.975]
intercept	0.0324	0.015	2.167	0.030	0.003	0.062
ar.L1	0.5509	0.108	5.102	0.000	0.339	0.763
ar.L2	0.2047	0.107	1.915	0.055	-0.005	0.414
ma.S.L12	-0.6353	0.139	-4.558	0.000	-0.909	-0.362
sigma2	0.0013	0.000	6.575	0.000	0.001	0.002

Forecasting with PMDARIMA (120): 3

model = pm.auto_arima(train, seasonal=True, m=12)

fit1 = pm.auto_arima(y, m=12, trace=True,

suppress_warnings=True)

Best model: ARIMA(2,1,1)(0,1,0)[12]

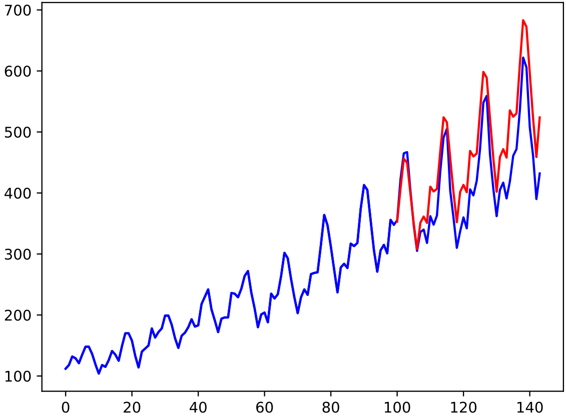
model.summary()

	coef	std err	z	P>|z|	[0.025	0.975]
intercept	0.0179	0.010	1.800	0.072	-0.002	0.037
ar.L1	0.6151	0.089	6.948	0.000	0.442	0.789
ar.L2	0.2366	0.092	2.559	0.010	0.055	0.418
ma.S.L12	-0.5564	0.120	-4.626	0.000	-0.792	-0.321
sigma2	0.0013	0.000	7.466	0.000	0.001	0.002

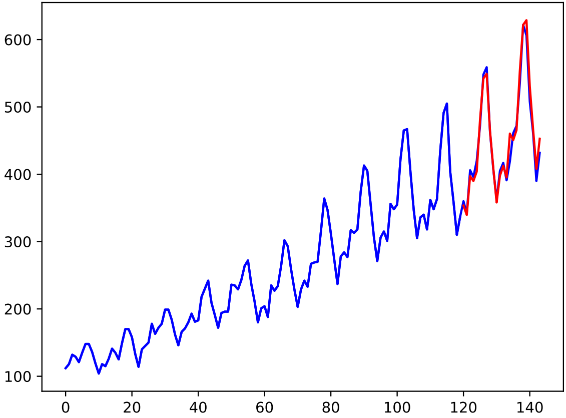
Forecasting with PMDARIMA: 4

```
#make forecasts
forecasts = model.predict(test.shape[0])
y = np.exp(y)
train = np.exp(train)
forecasts = np.exp(forecasts)
# Visualize the forecasts (blue=train, red=forecasts)
x = np.arange(y.shape[0])
plt.plot(x[:100], train, c='blue')
plt.plot(y, c = 'blue')
plt.plot(x[100:], forecasts, c='red')
plt.show()
```

Airline Data, Training = 100, Testing = 40



Airline Data, Training = 120, Testing = 20

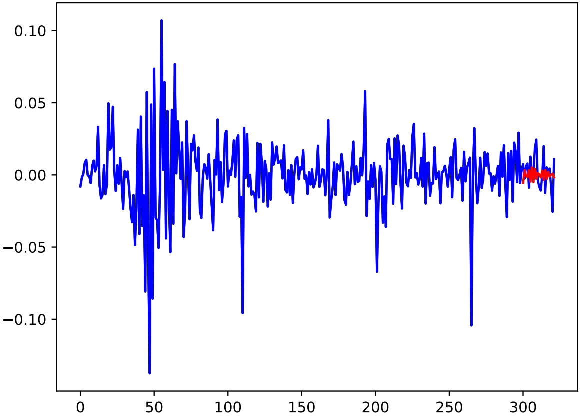


Forecasting IBM 1/1/20-4/14/21 with PMDARIMA

Best model: ARIMA(2,0,2)(0,0,0)[12]

	coef	std err	z	P>|z|	[0.025	0.975]
ar.L1	-1.7709	0.049	-36.153	0.000	-1.867	-1.675
ar.L2	-0.9062	0.046	-19.660	0.000	-0.997	-0.816
ma.L1	1.6792	0.070	23.899	0.000	1.542	1.817
ma.L2	0.7848	0.071	11.107	0.000	0.646	0.923
sigma2	0.0005	2.55e-05	21.408	0.000	0.000	0.001

IBM Data 1/1/20-4/14/21, Training = 300



ARCH/GARCH Modeling

Background: 1

- Thus far in the course, we have considered time series models, $\{X_t\}$, which are driven by a weakly stationary noise process, $\{\epsilon_t\}$, which, therefore, have constant variance.
- The implies that for models like AR, MA, ARMA and ARIMA, the time series also has constant variance.
- A stylized fact about most financial markets is that volatility is stochastic, and it tends to cluster into periods of low and high volatility.
- Consequently, standard time series models are not suitable for financial market data.
- Transformations like Box-Cox won't help as they cannot create volatility clusters.

Background: 2

- It is important to think about what sort of model we would want to introduce to create volatility clustering.
- Consider some of the graphs from early in the course - IBM, XOM, S&P 500 etc. The time series plots exhibit extremely different volatility patterns over time, and they tend to cluster.
- But the volatility clusters tend to be at the same time for all the stocks.
- This is indicative of a common factor that is impacting all of the stocks simultaneously, rather than the volatility patterns being created within the stocks themselves.

Background: 3

- GARCH models create volatility patterns "by themselves," i.e. it is built into their own stochastic model.
- Stochastic volatility models create an independent or correlated volatility process which then determines the instantaneous volatility of the stock(s).
- We will consider both in the coming days.

Background: 4

Some of the characteristics about volatility that have been noted in stock price behavior include:

- Volatility clustering
- Volatility is continuous, not a jump process
- Volatility does not diverge to infinity; in fact it is approximately stationary
- Volatility increases more following big price drops than after big price increases (leverage effect)

Background: 5

- For a weakly stationary time series model, $\{X_t,\ t \geq 0\}$
- If we define $\mu_t = E(X_t|\mathcal{F}_{t-1})$ and $\sigma_t^2 = Var(X_t|\mathcal{F}_{t-1})$, then for most models we have considered (AR, MA, ARMA, ARIMA), μ_t is non-constant and σ_t^2 is constant.
- For example, for an AR(1) process,

$$E(X_t|\mathcal{F}_{t-1}) = E(\phi_0 + \phi_1 X_{t-1} + \epsilon_t|\mathcal{F}_{t-1}) = \phi_0 + \phi_1 X_{t-1}$$

$$Var(X_t|\mathcal{F}_{t-1}) = Var(\phi_0 + \phi_1 X_{t-1} + \epsilon_t|\mathcal{F}_{t-1}) = Var(\epsilon_t) = \sigma_\epsilon^2.$$

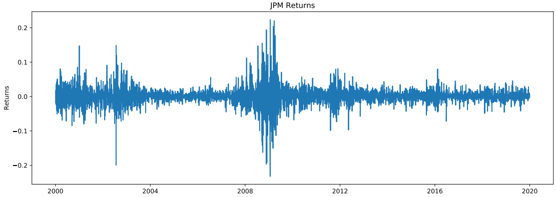
- The models we next develop are the opposite: μ_t is constant and σ_t^2 is non-constant.

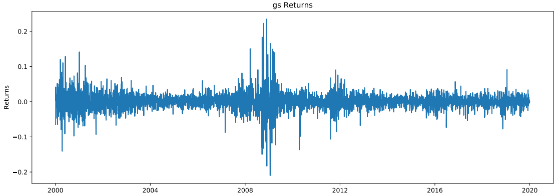
Background: 6

There are two distinct approaches to developing models with these properties:

- Deterministic Models: $Var(X_t|\mathcal{F}_{t-1})$ is a deterministic function. ARCH/GARCH models fall into this category
- Stochastic Models: $Var(X_t|\mathcal{F}_{t-1})$ is a stochastic process. Stochastic Volatility models (SV) fall into this category.

JPM and GS Returns





ARCH Models: 1

Following and quoting Ruppert p405, we discuss non-constant variance modeling.

- Suppose

$$Y_t = f(X_{1,t}, \cdots, X_{p,t}) + \epsilon_t.$$

  where $\{\epsilon_t\}$ is independent of the $\{X_{i,t},\ 1 \le i \le p,\ t \ge 0\}$ and has constant mean 0 and variance σ_ϵ^2.

- $E(Y_t|(X_{1,t}, \cdots, X_{p,t})) = f(X_{1,t}, \cdots, X_{p,t})$ and $Var(Y_t|(X_{1,t}, \cdots, X_{p,t})) = \sigma_\epsilon^2$

ARCH Models: 2

- It is possible to introduce conditional heteroscedasticity by letting
$$Var(Y_t|(X_{1,t},\cdots,X_{p,t})) = \sigma^2(X_{1,t},\cdots,X_{p,t}).$$
- The model
$$Y_t = f(X_{1,t},\cdots,X_{p,t}) + \epsilon_t\sigma(X_{1,t},\cdots,X_{p,t}),$$
where ϵ_t has, conditional on $(X_{1,t},\cdots,X_{p,t})$, mean 0 and variance 1, and $\sigma(\cdot)$ is non-negative since it is a standard deviation.
- If $\sigma(\cdot)$ is linear, then the coefficients need to be constrained so that it does not become negative for any values of the X variables.
- Generally, and in the GARCH case, the function $\sigma(\cdot)$ is non-negative and non-linear. These models are called variance function models, and GARCH will use the $\sqrt{\cdot}$ function

ARCH Models: 3

- The constant volatility models are called conditionally homoscedastic, whereas the models with stochastic volatility are called conditionally heteroscedastic.
- The first model, $\{a_t,\ t \geq 0\}$, is ARCH(1), Autoregressive Conditionally Heteroscedastic:

$$E(\epsilon_t | \epsilon_{t-1}, \epsilon_{t-2}, \cdots) = 0,$$

$$Var(\epsilon_t | \epsilon_{t-1}, \epsilon_{t-2}, \cdots) = 1,$$

$$a_t = \epsilon_t \sqrt{\omega + \alpha a_{t-1}^2},$$

a variance function model with $f = 0$ and $\sigma_t = \sqrt{\omega + \alpha a_{t-1}^2}$

ARCH Models: 4

- Ruppert's text takes $\{\epsilon_t,\ t \geq 0\}$ to be iid Gaussian white noise, but he indicates that more general assumptions are posssible.
- The further assumption $\omega > 0$ and $0 \leq \alpha < 1$ are made to ensure that the argument of the $\sqrt{\cdot}$ is positive.
- The $\{a_t^2,\ t \geq 0\}$ process is given by:

$$a_t^2 = \epsilon_t^2(\omega + \alpha a_{t-1}^2).$$

- This is somewhat similar to the AR(1) model except:
  - The variable is squared, a_t^2 rather than a_t, and
  - The noise process, ϵ_t is multiplicative rather than additive, and
  - The noise term, ϵ_t^2, is strictly positive and has mean $E(\epsilon_t^2) = 1$.

ARCH Models: 5

We first compute the conditional mean and variance of the a_t process. Let $\mathcal{F}_t$ be the sigma field generated by $\{a_s,\ s \leq t\}$, then

$$E(a_t|\mathcal{F}_{t-1}) = E(\epsilon_t\sqrt{\omega + \alpha a_{t-1}^2}|\mathcal{F}_{t-1}) = \sqrt{\omega + \alpha a_{t-1}^2}E(\epsilon_t|\mathcal{F}_{t-1}) = 0.$$

$$E(a_t|\mathcal{F}_{t-h})) = E(E(a_t|\mathcal{F}_{t-1})|\mathcal{F}_{t-h}) = E(0|\mathcal{F}_{t-h}) = 0.$$

$$\begin{aligned}
\sigma_t^2 &= Var(a_t|\mathcal{F}_{t-1}) \\
&= E(a_t^2|\mathcal{F}_{t-1}) \\
&= E((\omega + \alpha a_{t-1}^2)\epsilon_t^2|\mathcal{F}_{t-1}) \\
&= (\omega + \alpha a_{t-1}^2)E(\epsilon_t^2|\mathcal{F}_{t-1}) \\
&= \omega + \alpha a_{t-1}^2.
\end{aligned}$$

ARCH Models: 6

- Ruppert p408 "If a_{t-1} has an unusually large absolute value, then σ_t is larger than usual and so a_t is also expected to have an unusually large magnitude. This volatility propagates since when a_t has a large magnitude that makes σ_{t+1}^2 large, then a_{t+1} tends to be large in magnitude, and so on,"
- The same argument applies when a_{t-1} is unusually small.
- "Because of this behavior, unusual volatility in a_t tends to persist, though not forever. The conditional variance tends to revert to the unconditional variance provided that $\alpha < 1$, so the process is stationary with finite variance."

ARCH Models: 7

- Taking expectations we get $\gamma_a(0) = E(\sigma_t^2) = E(\omega + \alpha a_{t-1}^2) = \omega + \alpha\gamma_a(0)$. Assuming $0 \leq \alpha < 0$ and solving for $\gamma(0)$, we find

$$\gamma_a(0) = \frac{\omega}{1-\alpha}.$$

- We can find the covariance function for a_t using similar conditioning methods. For $h > 0$:

$$\begin{aligned}
\gamma_a(h) &= E(a_t a_{t-h}) \\
&= E(E(a_t a_{t-h} | \mathcal{F}_{t-h})) \\
&= E(a_{t-h} E(a_t | \mathcal{F}_{t-h})) \\
&= E(a_{t-h} * 0) \\
&= 0.
\end{aligned}$$

ARCH Models: 8

- It follows that the ACF of the ARCH(1) process a_t is $\rho_a(h) = 1$ if $h = 0$ and 0 if $h \neq 0$, a white noise process.
- The conditional variance expression $\sigma_t^2 = \omega + \alpha a_{t-1}^2$ gives insight into the behavior of ARCH(1) and GARCH models.
- If a_{t-1}^2 takes on an unusually large value, then $\sigma_t^2 = \omega + \alpha a_{t-1}^2$ will be unusually large, so we expect a_t to be large also (note though $\alpha < 1$). We expect this to create a cluster of large values for the a_t process, although the persistence depends importantly on the size of α.
- The same argument holds if a_{t-1}^2 takes on an unusually small value. There will tend to be a cluster of small volatility values.
- In both cases there is mean reversion to the mean value with α governing the speed of mean reversion.

AR(1)/ARCH(1)

We can combine the ARCH noise process with the AR time series model to achieve a process both of whose conditional mean and variance depend on the past.

- Suppose $\{a_t,\ t \in \mathcal{Z}\}$ is an ARCH(1) noise process where

$$a_t = \sqrt{\omega + \alpha a_t^2}\epsilon_t,$$

  where $\{\epsilon_t,\ t \in \mathcal{Z}\}$ consists of i.i.d. N(0,1) random variables, $\omega > 0$, $0 \leq \alpha < 1$, $|\beta| < 1$, and define

$$X_t = \mu + \beta(X_{t-1} - \mu) + a_t.$$

- $\rho_X(h) = \beta^{|h|}$ and $\rho_{a^2} = \alpha^{|h|}$.
- This model has non-constant conditional mean and variance.

ARCH(p)

- The order of the ARCH process can easily be increased from 1 to $p > 1$. Let
$$a_t = \sigma_t \epsilon_t,$$
where
$$\sigma_t = \sqrt{\omega + \sum_{i=1}^{p} \alpha_i a_{t-i}^2}.$$
- The ARCH(p) process is uncorrelated and has constant conditional and unconditional mean. Its unconditional variance is also constant, but the conditional variance is nonconstant.
- The ACF of a_t^2 is the same as the ACF of an AR(p) process.